

Transmission and Distribution Lines

1 Transmission Lines: Overhead vs Underground (7.1)

Overhead lines: bare conductors on poles/towers, supported by insulators.

- Cheaper to install and maintain, easy fault location, easy repair, good heat dissipation (higher current-carrying capacity for same size).
- Exposed to weather, lightning, wind, and interference; poorer aesthetics; needs more right-of-way (way-leave).

Underground cables: insulated conductors laid in the ground (direct buried, in ducts, or in pipes).

- Safer, unaffected by weather/lightning, better appearance, more reliable (no storm/wind faults), less interference with communication lines.
- Costlier (5–10 times overhead for same voltage), difficult fault location, poor heat dissipation (lower current rating), joints are weak points, higher charging current due to high capacitance.

Factor	Overhead	Underground
Cost	Low	High
Maintenance	Easy	Difficult
Fault location	Easy	Difficult
Appearance	Poor	Good
Public safety	Lower	Higher
Capacitance/charging current	Low	High
Life	Shorter	Longer

MCQ point: Underground cables have much higher capacitance than overhead lines $\Rightarrow$ larger charging current, more relevant at low/medium voltage distribution in cities.

2 Transmission Line Circuit Selection (7.2)

2.1 Necessity of High Voltage Transmission

For a given 3-phase power $P = \sqrt{3} V I \cos \phi$.

Formula:
$$I = \frac{P}{\sqrt{3} V \cos \phi}$$

- Increasing $V \Rightarrow$ decreases I for same power $\Rightarrow$ reduces $I^2 R$ (copper) losses, reduces conductor cross-section (less copper/aluminium), reduces % voltage drop.
- Disadvantage: higher V needs more insulation, larger clearances, bigger towers/insulator strings $\Rightarrow$ higher capital cost of insulation and supports.
- **Most economical voltage (Kelvin's law type argument):** chosen where the sum of annual cost of conductor + annual cost of insulation is minimized — cost of conductor decreases and cost of insulation increases as voltage increases, and total cost is minimized at an optimum voltage.

2.2 Kelvin's Law (Economical Conductor Size)

Total annual cost = annual charge on capital cost of conductor (area a) + annual cost of energy lost ($1/a$).

Formula: Most economical area is where: Annual interest and depreciation on conductor cost = Annual cost of energy loss

MCQ point: Kelvin's law gives the most economical cross-sectional area of a conductor, not voltage level directly.

2.3 Conductor Spacing

Governed by: sag/swing of conductors under wind, lightning protection, need to avoid clashing, and reduction of electrostatic stress & corona.

- Larger spacing $\Rightarrow$ increases inductance L , decreases capacitance C .
- Very close spacing $\Rightarrow$ higher electric stress $\Rightarrow$ more corona loss.

2.4 Insulating Materials

Porcelain (most common, high mechanical & dielectric strength), toughened glass (visible cracks on failure, used in some countries), and resin/polymer (composite, light, good for polluted/coastal areas).

2.5 Voltage Regulation and Efficiency of Short & Medium Lines

Voltage regulation (%VR): Formula: $\%VR = \frac{|V_S| - |V_R|}{|V_R|} \times 100$ where V_S = sending-end voltage, V_R = receiving-end voltage (both at same load; V_R usually held constant at rated value).

Transmission efficiency: Formula: $\eta = \frac{P_R}{P_S} = \frac{P_R}{P_R + \text{Line losses}} \times 100$

Short transmission line (length < 80 km, low/medium voltage): shunt capacitance neglected; only series R and X considered.

- Equivalent circuit: series impedance $Z = R + jX$ only.
- $V_S = V_R + IZ$ (phasor sum)
- ABCD constants: $A = 1$, $B = Z$, $C = 0$, $D = 1$

Medium transmission line (80–250 km): shunt capacitance is significant, but line is represented by lumped (not distributed) parameters.

- **Nominal- π model:** total capacitance split into two halves, placed at sending and receiving ends, with series impedance Z in the middle.

$$A = D = 1 + \frac{ZY}{2}, \quad B = Z, \quad C = Y \left(1 + \frac{ZY}{4}\right)$$

- **Nominal-T model:** total series impedance split into two halves with the full shunt capacitance Y lumped at the midpoint.

$$A = D = 1 + \frac{ZY}{2}, \quad B = Z \left(1 + \frac{ZY}{4}\right), \quad C = Y$$

- General relation: $V_S = AV_R + BI_R$, $I_S = CV_R + DI_R$, with $AD - BC = 1$.

MCQ point: Long lines (> 250 km) use distributed parameters and hyperbolic (ABCD) functions with characteristic impedance $Z_c = \sqrt{z/y}$ and propagation constant $\gamma = \sqrt{zy}$ — good to remember even though it is examined more under Power System Analysis.

3 Mechanical Design of Overhead Line (7.3)

3.1 Sag and Tension

For a conductor with supports at the same level: **Formula:** $S = \frac{wl^2}{8T}$ where w = weight of conductor per unit length, l = span length, T = tension in conductor.

For supports at different levels (height difference h), sag is measured separately from each support using the lowest point of the curve.

Effect of wind and ice loading: effective (resultant) weight becomes **Formula:** $w_t = \sqrt{(w + w_i)^2 + w_w^2}$ where w_i = weight of ice per unit length, w_w = wind force per unit length. Sag then acts along the slant direction:

$$S = \frac{w_t l^2}{8T}$$

- **Higher tension** $\Rightarrow$ lower sag, but increases mechanical stress on conductor and supports.
- Sag must be enough to allow for thermal expansion (temperature rise increases sag) and prevent excessive tension (which can snap the conductor).
- **Safety factor** of conductor: ratio of ultimate tensile strength to working tension (typically 2).

3.2 Supports and Cross Arms

- **Wooden poles** — cheap, used for low voltage/rural, short life, prone to decay.
- **Steel/RSJ poles** — stronger, used for medium voltage.
- **RCC poles** — long life, low maintenance, heavier, used widely for LT/HT distribution.
- **Steel towers (lattice)** — used for high/extra-high voltage transmission, strongest, longest spans.
- **Cross arms** hold the insulators/conductors; their length and spacing increase with voltage level (to maintain clearance and prevent flashover).

3.3 Conducting Materials

- **Copper:** highest conductivity, high tensile strength, costly, heavy — rarely used now for OH lines.
- **ACSR (Aluminium Conductor Steel Reinforced):** aluminium strands (conductivity) around a steel core (mechanical strength); most widely used, lighter and cheaper than copper for equal conductance.
- **AAAC / AAC / ACAR:** alternative aluminium-alloy conductors used depending on corrosion, strength, and conductivity needs.

3.4 Types of Insulators

- **Pin type:** used up to about 33 kV; conductor tied on top; single-piece or multi-part porcelain shells.

- **Suspension (disc) type:** conductor hangs below the string; used for higher voltages by adding more discs in series; standard for transmission lines.
- **Strain insulators:** used at dead ends, sharp corners, and river crossings where line is subjected to full tension.
- **Shackle insulators:** used in low-voltage distribution lines, mounted horizontally or vertically on pole.
- **Stay/egg insulators:** inserted in the stay wire for electrical isolation.

3.5 String Efficiency

Suspension insulator string = several discs in series; each disc has self-capacitance C plus stray/shunt capacitance C' to earth/tower ($k = C'/C$). Because of shunt capacitance leakage, voltage across each disc is **not equal** — the disc nearest the conductor carries the highest voltage.

Formula: String efficiency = $\frac{\text{Voltage across whole string}}{n \times \text{voltage across disc nearest conductor}} \times 100\%$ where n = number of discs.

- String efficiency is always less than 100% and decreases as n (number of discs) increases.
- **Methods to improve string efficiency:**
 1. Using a **longer cross-arm** (reduces the ratio $k = C'/C$).
 2. **Grading of insulators** — using discs of different (increasing) capacitance toward the line conductor so voltage divides more equally.
 3. **Static/guard (grading) ring** — a metal ring connected to the conductor end that introduces an equalizing capacitance, forcing the shunt capacitive currents to balance so voltage distributes uniformly.

MCQ point: String efficiency = 100% only in the ideal case of zero stray capacitance ($k = 0$), i.e. equal voltage across every disc.

3.6 Jumpers and Vibration Dampers

- **Jumper:** a short piece of conductor used at tension/strain towers to electrically connect the conductors of adjacent spans (since the main conductor is dead-ended at each side of the tower).
- **Vibration dampers (e.g. Stockbridge damper):** clamped to the conductor near supports to absorb energy from wind-induced (aeolian) vibrations, preventing fatigue failure of strands at the point of support.

4 Electrical Loads (7.4)

4.1 Types of Loads

Domestic, commercial, industrial, agricultural, municipal (street lighting, water supply), traction (railways).

4.2 Load Characteristics — Key Definitions and Formulas

Formula: Demand factor = $\frac{\text{Maximum demand}}{\text{Connected load}}$ (always ≤ 1)

Formula: Average load = $\frac{\text{Total energy (units) consumed}}{\text{Time period (hours)}}$

Formula: Load factor = $\frac{\text{Average load}}{\text{Maximum demand}}$ (always ≤ 1) Higher load factor $\Rightarrow$ load is more uniform $\Rightarrow$ better utilization of generating plant $\Rightarrow$ lower cost per unit generated.

Formula: Diversity factor = $\frac{\text{Sum of individual maximum demands}}{\text{Maximum demand of the system}}$ (always ≥ 1) Higher diversity factor $\Rightarrow$ individual peaks don't occur together $\Rightarrow$ smaller plant capacity needed $\Rightarrow$ lower capital cost.

Formula: Plant capacity factor = $\frac{\text{Average demand}}{\text{Plant (installed) capacity}}$

Formula: Plant use factor = $\frac{\text{Units generated}}{\text{Plant capacity} \times \text{hours plant was actually in operation}}$

MCQ point: Load factor relates to how the same load varies over time; Diversity factor relates to different loads not peaking together.

4.3 Load Curve and Load Forecasting

- **Load curve:** plot of load (kW) vs time over a day/year — used to determine peak load, base load, and load factor.
- **Load duration curve:** same data rearranged in descending order of magnitude, showing the duration for which a given load persists.
- **Load forecasting:** predicting future demand using trend/extrapolation methods, correlation with population/GDP growth; classified as short-term (hours to a year, for operation planning) and long-term (years, for system expansion planning).

5 Distribution Systems (7.5)

5.1 Primary Distribution System

Carries power at higher voltage (e.g. 11 kV / 33 kV typical) from substation to distribution transformers.

- **Radial system:** power flows in one direction only, single feeder branching out; simplest and cheapest, but a fault anywhere interrupts supply to all consumers beyond that point — **low reliability**.
- **Ring main system:** feeder forms a closed loop returning to the substation, so every point can be fed from two directions; a single fault can be isolated without cutting off supply — **better reliability, moderately costly**.
- **Interconnected system:** the ring/network is fed from more than one substation/generating station; **most reliable**, continuity of supply maintained even if one source fails, but most costly and needs more complex protection coordination.

5.2 Secondary Distribution System

Carries power at utilization (low) voltage from distribution transformer to consumers.

- **Three-phase, four-wire system** (three phases + neutral, e.g. 400/230 V): supplies both 3-phase loads (motors) and single-phase loads (lighting, sockets) using phase-to-neutral voltage; most

common for mixed domestic/industrial areas.

- **Single-phase, two-wire system:** one phase + neutral; used for small, purely single-phase loads such as isolated rural consumers.

5.3 Selection of Supports and Conductors in Secondary Distribution

Based on: expected load current (ampacity), permissible voltage drop, mechanical strength required for span/sag, and environmental exposure (corrosion, pollution).

5.4 Stay Wire (Guy Wire)

Used to counteract the unbalanced pull on a pole at dead ends, corners, and sharp angle points (where conductor tension is not balanced on both sides).

- Anchored to the ground using a stay rod and stay plate/block; an **egg/stay insulator** is inserted in the wire to break earth continuity for safety.
- **Types:** normal (single) stay, **span/horizontal (H-type) stay** (used when ground anchoring space is unavailable, tied to an adjacent pole instead), and anti-theft/anti-climbing stay arrangements.

6 Voltage Regulation and Power Factor Correction (7.6)

6.1 Voltage Regulation in Distribution System

Methods to maintain voltage within limits despite varying load:

- **Tap-changing transformers:** off-load tap changers (changed only when de-energized) and on-load tap changers (OLTC, changed under load without interruption).
- **Booster transformers:** injected in series with the feeder to add/subtract a small voltage.
- **Shunt capacitors:** supply reactive power locally, reduce line current and voltage drop.
- **Synchronous condensers:** over-excited synchronous motor (no mechanical load) supplying reactive power, can be smoothly varied, but costly and needs maintenance.
- **Static VAR compensators (SVC):** fast, solid-state reactive power control.

Recall: **Formula:** $\%VR = \frac{V_S - V_R}{V_R} \times 100$

6.2 Power Factor Correction

Why low power factor is undesirable:

- For fixed real power P , low $\cos \phi \Rightarrow$ higher current $I = P/(V \cos \phi) \Rightarrow$ higher I^2R losses, larger conductor size needed, poorer voltage regulation, and reduced effective kVA capacity of generators/transformers/lines.
- Utilities often impose a **penalty** on consumers with poor power factor.

Methods of power factor improvement:

1. **Static capacitors** (shunt-connected) — most common, supply leading reactive power to offset lagging reactive power of inductive loads (motors); low maintenance, but fixed (or steps).

2. **Synchronous condensers** — smoothly adjustable, can also help with voltage support, but costly and has running losses.
3. **Phase advancers** — used with induction motors specifically, mounted on motor shaft to supply the exciting mmf, improving pf of the individual motor.

Capacitor rating required to improve power factor: Formula: $Q_C = P (\tan \phi_1 - \tan \phi_2)$ where P = active power (kW), ϕ_1 = original power factor angle, ϕ_2 = desired (improved) power factor angle, Q_C in kVAR.

MCQ point: Improving power factor from lagging toward unity always reduces line current and I^2R losses for the same real power delivered.